

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 10%

Code of Conduct:

I V. Thanusri (192472265) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Thanusri

Rubrics for Evaluation

<i>Criteria</i>	<i>Weightage</i>	<i>Level 4 – Exemplary</i>	<i>Level 3 – Proficient</i>	<i>Level 2 – Developing</i>	<i>Level 1 – Beginning</i>	<i>Marks(100)</i>
<i>Understanding of Concepts</i>	<i>25%</i>	<i>Demonstrates thorough, accurate understanding of concepts</i>	<i>Good understanding with minor gaps</i>	<i>Basic understanding with some misconceptions</i>	<i>Poor understanding; major misconceptions</i>	
<i>Explanation</i>	<i>25%</i>	<i>Detailed, well-explained answers with relevant examples</i>	<i>Clear explanation with adequate details</i>	<i>Limited explanation; lacks depth</i>	<i>Poor or unclear explanation</i>	
<i>Application</i>	<i>20%</i>	<i>Effectively applies concepts with strong</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>	

		<i>analytical ability</i>				
<i>Organization</i>	<i>15%</i>	<i>Well-structured answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization, incoherent answers</i>	
<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>	
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>	

1. An image contains high-frequency noise. Explain and justify the appropriate frequency domain filter, including its working and impact on image quality.

Answer

Introduction

High-frequency noise consists of sudden intensity variations caused by sensor noise or transmission errors. Since noise occupies the high-frequency region, a Low Pass Filter (LPF) is the most suitable frequency-domain filter.

Appropriate Filter

Ideal Low Pass Filter (ILPF), Butterworth Low Pass Filter (BLPF), or Gaussian Low Pass Filter (GLPF).

Among these, Gaussian Low Pass Filter is preferred because it produces smooth results without ringing artifacts.

Working

1. Convert the image from the spatial domain to the frequency domain using the Fourier Transform (FFT).

2. Apply the low-pass filter to allow only low-frequency components.
3. High-frequency components containing noise are attenuated.
4. Perform the Inverse Fourier Transform (IFFT) to reconstruct the filtered image.

The Gaussian LPF is given by:

$$H(u,v) = e^{-\frac{D^2(u,v)}{D_0^2}}$$

where:

- $D(u,v)$ = distance from the frequency center
- D_0 = cutoff frequency

Impact on Image Quality

Advantages

- Removes high-frequency noise effectively.
- Produces smooth and natural-looking images.
- Preserves important low-frequency information.

Disadvantages

- Slight blurring of edges.
- Fine details may be lost if the cutoff frequency is too low.

Justification

Since high-frequency noise occupies the high-frequency spectrum, suppressing these frequencies removes noise while preserving the main image content.

Conclusion

Gaussian Low Pass Filtering provides effective noise reduction with minimal artifacts, making it the preferred choice.

2. You need to preserve low-frequency components while removing noise. Describe and apply a suitable filtering technique, explaining how it achieves the desired effect.

Answer

Introduction

Most useful image information is present in low-frequency components, whereas random noise mainly appears at high frequencies.

Suitable Technique

Gaussian Low Pass Filter (GLPF)

Working

1. Apply FFT to convert the image into the frequency domain.
2. Multiply the spectrum with the Gaussian LPF.
3. High-frequency noise is reduced while low-frequency information is retained.
4. Apply IFFT to reconstruct the image.

Transfer function:

$$H(u,v) = e^{-\frac{D^2(u,v)}{2D_0^2}}$$

Why it Works

- Smoothly reduces high frequencies.
- Avoids sudden cutoff.
- Preserves image brightness and major structures.

Advantages

- Excellent noise reduction.
- No ringing artifacts.
- Better visual quality than Ideal LPF.

Limitations

- Slight loss of edge sharpness.
- Excessive filtering causes image blur.

Conclusion

Gaussian Low Pass Filtering effectively preserves important low-frequency information while removing unwanted noise.

3. An image requires edge enhancement in the frequency domain. Explain and justify the filtering approach used, describing its effect on frequency components.

Answer

Introduction

Edges represent rapid intensity changes and are contained in high-frequency components.

Suitable Filter

High Pass Filter (HPF)

Examples:

- Ideal HPF
- Butterworth HPF
- Gaussian HPF

Working

1. Convert the image using FFT.
2. Apply HPF.
3. Low frequencies are suppressed.
4. High frequencies corresponding to edges are enhanced.
5. Perform IFFT.

Gaussian HPF:

$$H(u,v) = 1 - e^{-\frac{D^2(u,v)}{2D_0^2}}$$

Effect on Frequency Components

- Removes low-frequency background.
- Enhances high-frequency edges.
- Increases image sharpness.

Advantages

- Highlights boundaries.

- Improves image details.
- Sharpens blurred images.

Limitations

- Noise may also be amplified.
- May require smoothing before enhancement.

Conclusion

High Pass Filtering is ideal for edge enhancement because edges correspond to high-frequency information.

4. A smooth version of an image is required. Describe and justify the frequency domain technique used for smoothing, explaining its operation and results.

Answer

Introduction

Image smoothing removes noise and small intensity variations.

Suitable Technique

Low Pass Filtering (Gaussian LPF preferred)

Operation

1. Convert image into frequency domain.
2. Apply LPF.
3. Remove high-frequency components.
4. Reconstruct image using IFFT.

Results

- Smooth image.
- Reduced noise.
- Less texture variation.
- Slight edge blurring.

Advantages

- Effective denoising.
- Preserves overall image appearance.
- Easy implementation.

Disadvantages

- Fine details may disappear.
- Sharp edges become blurred.

Justification

Since noise is concentrated in high frequencies, LPF smooths the image by suppressing these components.

Conclusion

Frequency-domain low-pass filtering provides efficient image smoothing with acceptable loss of detail.

5. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.

Answer

Introduction

When an object is significantly brighter than the background, Threshold Segmentation is the simplest and most effective method.

Suitable Technique

Global Thresholding

Working

1. Choose a threshold value TTT.
2. Compare every pixel with TTT.
3. Pixels greater than TTT become object.
4. Remaining pixels become background.

Formula:

$$g(x,y)=\begin{cases} 1, & f(x,y) > T \\ 0, & f(x,y) \leq T \end{cases}$$

Advantages

- Simple.
- Fast.
- Accurate for high contrast images.

Limitations

- Poor performance under uneven illumination.
- Not suitable for overlapping intensities.

Effectiveness

Since the object is much brighter, thresholding clearly separates foreground from background.

Conclusion

Threshold segmentation is the most suitable technique when object intensity is distinctly higher than the background.

6. You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.

Answer

Introduction

Segmentation based on intensity divides an image into object and background.

Suitable Method

Thresholding (Global or Otsu's Thresholding)

Process

1. Analyze image histogram.
2. Select threshold.
3. Convert image into binary form.

4. White pixels represent objects.
5. Black pixels represent background.

Advantages

- Simple implementation.
- Fast computation.
- Effective for bimodal histograms.

Limitations

- Sensitive to lighting.
- Requires preprocessing in noisy images.

Application

Widely used in:

- Medical imaging
- Document processing
- Industrial inspection

Conclusion

Thresholding is an efficient segmentation technique for separating foreground and background using intensity values.

7. An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.

Answer

Introduction

Edge detection identifies object boundaries by detecting sudden intensity changes.

Suitable Method

Canny Edge Detector

Working

1. Smooth image using Gaussian filter.

2. Compute gradients using Sobel operator.
3. Apply Non-Maximum Suppression.
4. Use Double Thresholding.
5. Perform Edge Tracking by Hysteresis.

Advantages

- High detection accuracy.
- Good localization.
- Low false edge detection.
- Noise resistant.

Limitations

- Computationally expensive.
- Requires threshold selection.

Effectiveness

Produces thin, continuous, and accurate edges while minimizing noise.

Conclusion

Canny Edge Detection is considered the best edge detection algorithm because of its accuracy and robustness.

8. Adjacent pixels have similar intensity values. Describe and apply a segmentation technique to group such pixels, explaining the concept and reasoning.

Answer

Introduction

When neighboring pixels have similar intensity values, Region Growing is an appropriate segmentation technique.

Suitable Technique

Region Growing

Working

1. Select seed point(s).
2. Compare neighboring pixels.
3. Add pixels with similar intensity.
4. Continue until similarity condition fails.

Advantages

- Produces connected regions.
- Good boundary detection.
- Simple implementation.

Limitations

- Sensitive to seed selection.
- Noise may affect results.

Reasoning

Pixels belonging to the same object usually have similar intensity and are spatially connected.

Applications

- Medical image segmentation.
- Satellite image analysis.
- Object extraction.

Conclusion

Region Growing effectively groups adjacent pixels with similar properties into meaningful regions.

9. Detected edges in an image appear broken.

Explain and justify a method to improve edge continuity, including its role in post-processing.

Answer

Introduction

Broken edges reduce the quality of object detection and segmentation.

Suitable Method

Morphological Closing

Working

1. Apply dilation to connect nearby edges.
2. Apply erosion to restore edge thickness.
3. Small gaps are filled.
4. Continuous boundaries are obtained.

Advantages

- Connects broken edges.
- Removes small holes.

Limitations

- Large structuring elements may merge nearby objects.
- May slightly alter boundary shape.

Role in Post-processing

Morphological Closing is commonly applied after edge detection (e.g., Canny) to improve edge continuity and prepare images for segmentation or object recognition.

Conclusion

Morphological Closing effectively repairs broken edges and improves the quality of edge-based image analysis.

10. You need to extract object outlines from an image. Describe and apply an appropriate technique, explaining how it identifies boundaries effectively.

Answer

Introduction

Object outline extraction identifies the external boundaries of objects.

Suitable Technique

Boundary Extraction using Morphological Operations

Formula:

$$\text{Boundary} = A - (A \ominus B)$$

where:

- A = original object
- B = structuring element
- $\ominus$ = erosion

Working

1. Perform erosion on the binary image.
2. Subtract the eroded image from the original image.
3. Remaining pixels represent object boundaries.

Advantages

- Simple.
- Accurate boundary extraction.
- Preserves object shape.

Limitations

- Requires binary image.
- Depends on structuring element size.

Applications

- Shape analysis.
- Object recognition.
- Medical imaging.
- Industrial inspection.

Conclusion

Morphological Boundary Extraction accurately identifies object outlines by removing interior pixels and retaining only boundary pixels.